

Federated Learning over wireless links in smart City

Artificial Intelligence (AI) is transforming the way cities operate, making them smarter, more efficient, and more sustainable. AI offers a wide range of applications in smart cities, including traffic management, service monitoring, energy optimization, public safety, and more. By integrating AI into urban systems, cities can address complex challenges and improve the overall quality of life for their residents. Neural Network (NN) schemes are becoming increasingly popular across various fields and applications. With the vast amounts of data generated by wireless devices, distributed learning techniques are gaining significant traction in multiple industry sectors. Among these innovative approaches, Federated Learning (FL) stands out as a promising method that enables multiple edge devices to collaboratively train a shared model while preserving data privacy. In FL, edge devices use their local data to update the model by computing the gradient of the loss function. They then send either the gradients or the locally updated model to a central server. The server aggregates the received updates and distributes the improved model back to the devices, repeating the process iteratively.

The goal of this project is to explore the application of FL in smart city scenarios and implement an FL framework using a simple labeled dataset. We will evaluate the performance of the FL system in terms of accuracy and convergence time. Additionally, we will analyze the energy consumption and processing complexity to assess the cost implications of deploying FL in smart city applications. This investigation will provide valuable insights into the feasibility and efficiency of FL for enhancing smart city systems.

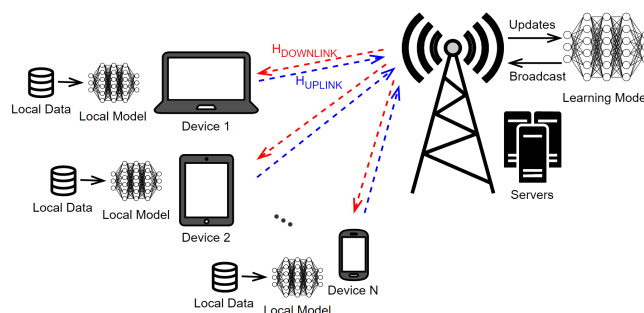


Figure 1: Federated learning over wireless networks.

Topics: Optimization theory, Federated Learning, Neural Network, gradient descent

Software to use: Matlab or Python